

XAI Experience Quality (XEQ) Scale Handbook

Practical guide for applying and interpreting the 3-factor XEQ Scale

Updated for Utility, Satisfaction, and Effectiveness

V.2.0.0

Introduction

The XAI Experience Quality (XEQ) Scale was developed to help researchers and practitioners evaluate how users experience AI explanations — especially in interactive, multi-step settings. While the original research provides the theoretical foundation and validation, this handbook serves as a practical guide to applying the scale.

Our aim is to make the XEQ Scale easy to use in real-world projects. This handbook summarises the essentials: what the scale measures, how to apply it, and how to interpret the results. Where relevant, we point to further resources for those who wish to explore the research in more depth.

Construction of the XEQ

The XEQ Scale was developed to assess the quality of user experiences with explainable AI systems, particularly those involving interactive, multi-shot explanations. The construction process followed established practices from psychometric theory to ensure both rigour and practical value.

An initial pool of 40 questionnaire items was compiled through a targeted literature review and expert brainstorming. These items were sourced from existing evaluation tools in XAI and HCI and extended with new items to reflect the interactive and user-centred nature of XAI experiences.

The item pool was refined through a two-stage expert review process and then evaluated in a content validity study involving 13 XAI researchers and practitioners. Based on expert feedback, the scale was reduced to 18 items, grouped into three key factors: Utility, Satisfaction, and Effectiveness.

A large-scale pilot study with 238 participants was conducted to validate the scale. Statistical analyses confirmed strong internal consistency (Cronbach's $\alpha = 0.9562$), discriminant validity, and test-retest reliability. Factor analysis identified an overarching XAI Experience Quality factor and supported a three-factor structure comprising Utility, Satisfaction, and Effectiveness.

For detailed information on the construction and validation of the XEQ Scale, see: Wijekoon, Anjana, Nirmalie Wiratunga, David Corsar, Kyle Martin, Ikechukwu Nkisi-Orji, Craig Pirie, Belen Díaz-Agudo, and Derek Bridge. 2026. "XEQ Scale for Evaluating XAI Experience Quality." *ACM Transactions on Interactive Intelligent Systems* 37, 4, Article 111 (June 2026), 30 pages.

Scale Structure

The XEQ Scale contains 18 items across 3 factors. Each factor reflects a distinct aspect of user experience when interacting with explainable AI (XAI) systems:

- **Utility (6 items):** Measures the contribution of the experience towards task completion, including confidence, decision support, role fit, timeliness, and task completion.
- **Satisfaction (5 items):** Captures the degree to which the experience supports the achievement of XAI goals, including reliability, expectations, requirements, perceived system performance, and overall satisfaction.
- **Effectiveness (7 items):** Captures the extent to which the experience resolves explanation needs and improves understanding through consistency, clarity, trust calibration, sensible progression, sufficient detail, and complete answers.

All items are rated on a 5-point Likert scale, from Strongly Disagree to Strongly Agree. These factors allow for both an overall XAI Experience Quality (XEQ) score and more detailed insights through per-factor scores.

Can I Remove Items?

You **cannot remove individual items from the questionnaire**, as each one contributes to the reliability of its corresponding XEQ factor. In practice, teams may focus interpretation on a specific factor or administer the scale to a single stakeholder group, but any factor-level adaptation should be reported clearly so results are not confused with the full 18-item XEQ score.

Typical Application Scenarios

Test if a system has sufficient quality

This scenario uses XEQ to assess the overall quality of a system. Factor scores across Utility, Satisfaction, and Effectiveness that are roughly equal and above 3 indicate that the system provides a balanced experience, with no major weaknesses across the factors. Such a pattern suggests that the product is likely to meet user expectations broadly and consistently, making it a suitable candidate for wider deployment or continued use. However, there is still room for improvement.

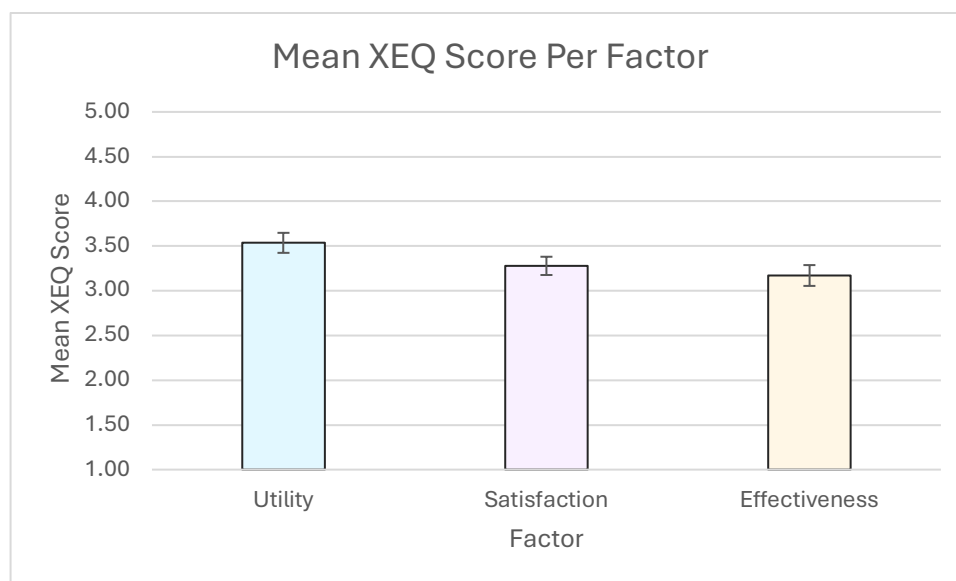


Figure 1: Example factor scores

Better personalise explanation experiences

This scenario uses XEQ to identify differences in experience quality across stakeholder groups. Stakeholder scores may reveal, for example, that one role rates Effectiveness or Utility much lower than others. This suggests that the current system or explanation approach does not meet that group's needs, either because the explanations do not support their task or because they do not resolve their explanation needs. These insights highlight the importance of personalising explanations based on role-specific expectations and background knowledge, ensuring that all users—not just some—can benefit meaningfully from the system.

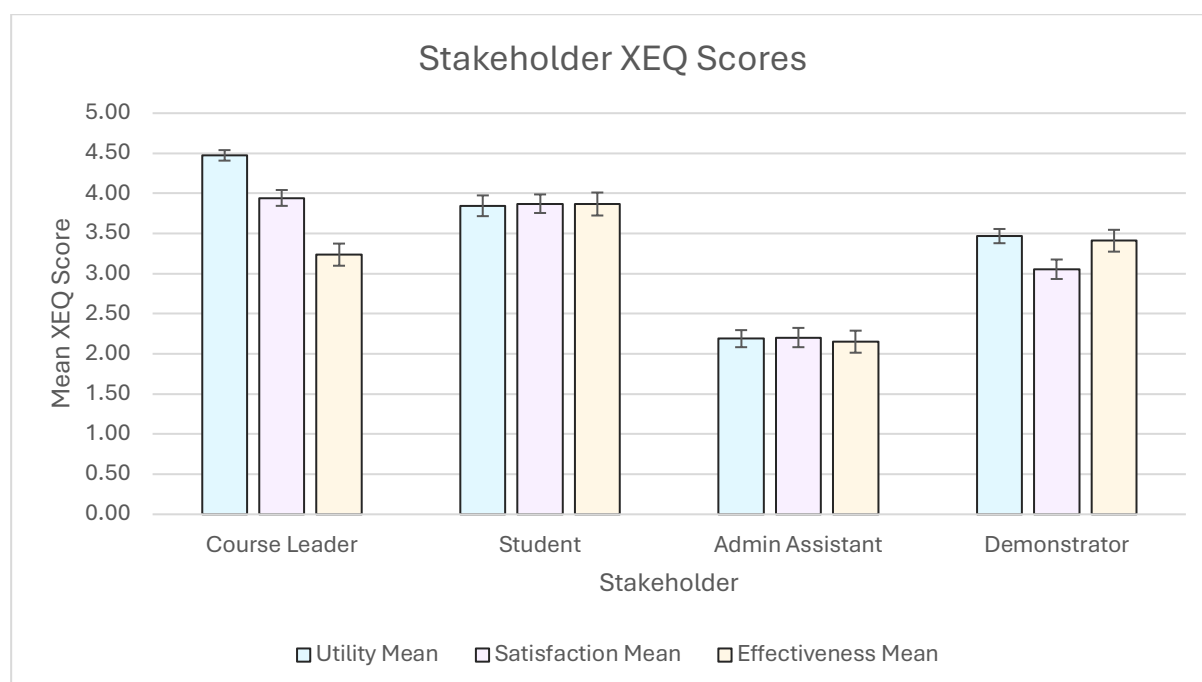


Figure 2: Example stakeholder scores

Compare competing XAI systems

XEQ can be used to compare the user experience quality of two or more AI systems performing similar tasks—for example, evaluating whether a new version of a model offers clearer, more useful, or more satisfying explanations than an existing one. By collecting XEQ responses for each system separately, you can compare scores across factors and stakeholder groups to identify which system delivers a better overall experience or meets specific user needs more effectively. While the current tool supports analysis of one system at a time, we plan to release a comparison sheet in the near future to make side-by-side evaluations easier and more structured.

Benchmarking

XEQ is designed to support **benchmarking** by enabling organisations to track experience quality over time or compare results across systems and domains. Although this functionality is conceptually supported, **no benchmark data is available yet**, as the tool has only just been released. As more users adopt XEQ and share results, we plan to develop a **benchmark dataset** to allow comparisons against typical experience scores. For now, you can begin building your own internal benchmarks by applying XEQ consistently across projects and tracking changes over time.

Translation to Other Languages

At present, the XEQ Scale is only available in English. However, we are actively working on translating it into other languages to make it more accessible to a global audience. Translations will be added as they become available, and we welcome community contributions. If you are interested in helping with a translation, please get in touch.

XEQ Hardcopy

A **printable hardcopy version** of the XEQ questionnaire is available for download from the [XEQ website](#). This format is ideal for use in interviews, workshops, or situations where digital access is limited. Responses collected on paper can later be entered into the Excel tool for analysis.

Online Application

The online version of the XEQ tool is designed for **individual use**—it allows a single user to complete the questionnaire and reflect on their experience. Once completed, users can **print out their responses to each item**, which can then be manually entered into the XEQ Excel tool for group-level analysis and visualisation. This is especially helpful for workshops, pilot testing, or gathering initial feedback. In future, we plan to develop an **online results collation tool** to make it easier to collect and analyse responses from multiple users automatically.

How to Use the Excel Tool?

The XEQ Excel Tool is designed to simplify the process of scoring and analysing responses to the XAI Experience Quality (XEQ) Scale. To use it, simply enter participant responses into the 'Data' sheet and define the stakeholders who you are interacting with your AI system in the 'Stakeholders' sheet within the file XEQ_Data_Analysis_Tool.xlsx.

All key metrics — including stakeholder scores, factor scores, and the overall system XEQ score — are automatically calculated. You do not need to perform any manual analysis or write formulas.

The example illustrates how XEQ scores can be used to evaluate perceptions of a single system across different stakeholder groups. While the current tool supports rich analysis of one system at a time, **it does not yet support direct comparisons between multiple systems or versions** (e.g., comparing version A vs. version B).

A separate comparison tool—**XEQ_Compare_Systems.xlsx**—is **planned for future release** to support such analyses. In the meantime, users can still manually compare outputs by placing results side by side and noting differences in scores and stakeholder patterns.

Please make sure you are using the **latest version** of the XEQ Excel Tool, available at www.xeqscale.org, as it is actively maintained and improved based on ongoing research and user feedback.

How to Interpret the Data

How do I interpret the scores?

The XEQ tool provides three layers of results to help you understand user experience from different perspectives:

- **Stakeholder scores:** These show how each group of users rated their experience. For example, if students give high scores across all three factors but admin staff score much lower, it suggests the product works well for learners but may not meet the needs of administrative roles. This helps identify whose experience is strong and where improvements may be needed.
- **System score:** This is the overall average across stakeholder groups and XEQ factors. For instance, a system score of 4.2 indicates that, on average, users had a very positive experience. If this number is closer to 3.0, the product may still need refinement. Think of it as a high-level snapshot of quality across the board.
- **Factor scores (Utility, Satisfaction, Effectiveness):** These tell you how the system performs in each of the three XEQ factors. For example, if Utility scores are high (e.g., 4.5), users found the experience supportive of their tasks and decisions. If Effectiveness scores are low (e.g., 2.8), the

explanations may not be clear, coherent, detailed, or complete enough to resolve users' explanation needs. This highlights specific areas where experience can be improved.

Taken together, these scores help you answer:

- *Who is having a good (or poor) experience?* (Stakeholder scores)
- *How good is the overall experience?* (System score)
- *Which aspects of the experience stand out or fall short?* (Factor scores)

As more projects use the XEQ tool, we plan to develop a **benchmark dataset** to support comparisons. This will allow you to see how your scores compare to typical ranges in similar systems, making interpretation even more meaningful and grounded in real-world performance. **For now, a simple rule of thumb is to interpret scores below 3 as needing improvement, and scores above 3 as acceptable or positive.**

What do the error bars mean?

In the XEQ tool, each mean score is shown with an accompanying **error bar**, which visually represents the **confidence interval** for that score. By default, this is set to **95%**, but users can adjust the confidence level based on their needs.

Think of it this way: if you were to repeat the same evaluation many times under the same conditions, small variations would naturally occur due to randomness. The error bar marks the range in which **95% of those repeated mean scores** would be expected to fall. In other words, it shows how precisely your result reflects the underlying truth.

Two main factors influence the size of the error bar:

- **Sample size:** Larger participant numbers tend to reduce the width of the interval.
- **Response consistency:** When participants agree more closely in their answers, the interval also becomes narrower.

Wider error bars suggest greater uncertainty in your measurements, meaning you should be more cautious in interpreting the results. Often, a large interval is a sign of too few responses. In such cases, gathering more data is advisable to improve the reliability of your findings.

How to interpret the standard deviation

Standard deviation is a measure of how much individual responses vary around the average score. A **low standard deviation** means that most people gave similar ratings, indicating agreement or consistency in their experience. A **high standard deviation**, on the other hand, suggests that responses were more spread out—some rated the experience highly, while others rated it much lower. This can be a sign of differing expectations, inconsistent performance, or variation in how different users interacted with the product.

Frequently Asked Questions (FAQs)

For answers to common questions and tips on getting started, please check the FAQ section on our website. As the XEQ tool has just been released, the FAQ is still being developed—but we'll be adding new entries based on user feedback and common queries. Stay tuned!